TerminalX

Suraksha Saathi

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Ensuring Safety, Independence & Rapid Response in Everyday Life

QUANTUM ARENA 1.0

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Problem Statement Title: Fall Detection Device for Elderly

College: Vishwakarma Institute of Technology, Pune

Team Name: TerminalX

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Problems with Current Emergency Systems

Delayed Emergency Response

Emergencies like falls often go unnoticed, especially indoors, leading to delayed medical assistance.

No Indoor Monitoring

Most systems rely on GPS and fail to monitor risks inside homes like bathroom falls or inactivity.

Lack of Automatic Detection

Elderly individuals may be unconscious or unable to call for help after a fall or medical event.

Lack of Assistive Navigation

Visually impaired individuals face difficulty navigating indoor environments safely without assistance.

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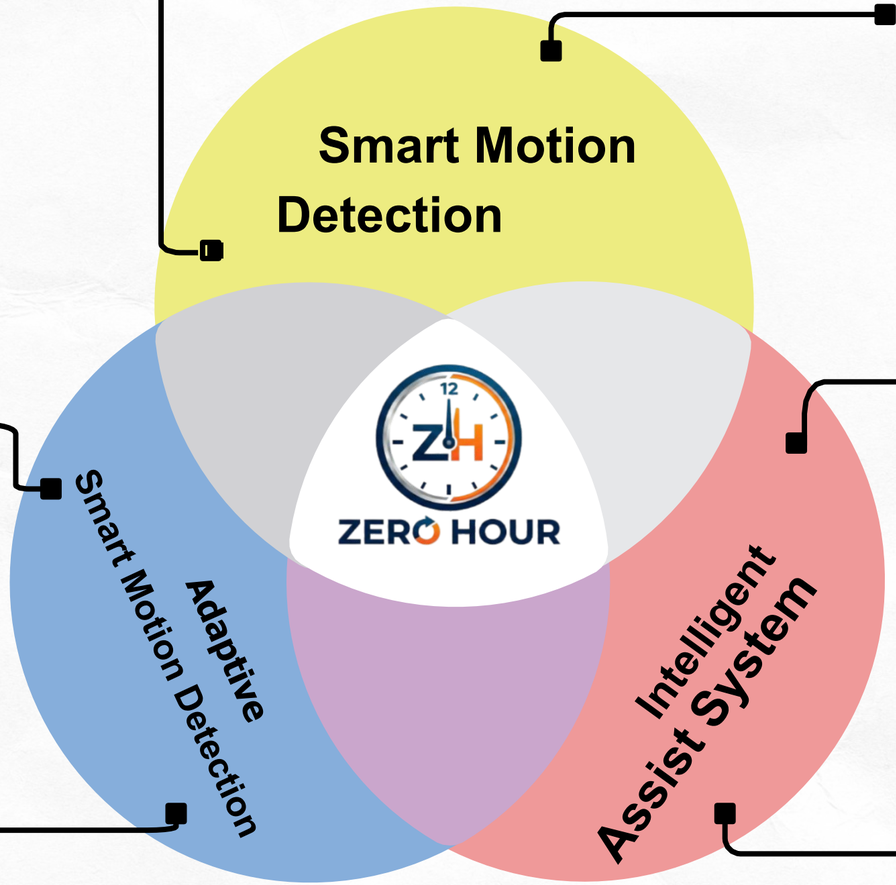
How does our system solve this?

Smart Fall Detection:
AI classifies accident as minor, moderate or severe.

Voice Assistance:
System provides audio guidance and emergency prompts.



Indoor Risk Detection:
Detects bathroom inactivity, night wandering and abnormal behavior..

Real Time GPS-Tracking:
Shares live location with caregivers during emergencies.

Emergency Alert System:
Sends instant alerts with location and health details.

User Health Profile:
Stores medical data for quick response during emergencies..

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Features

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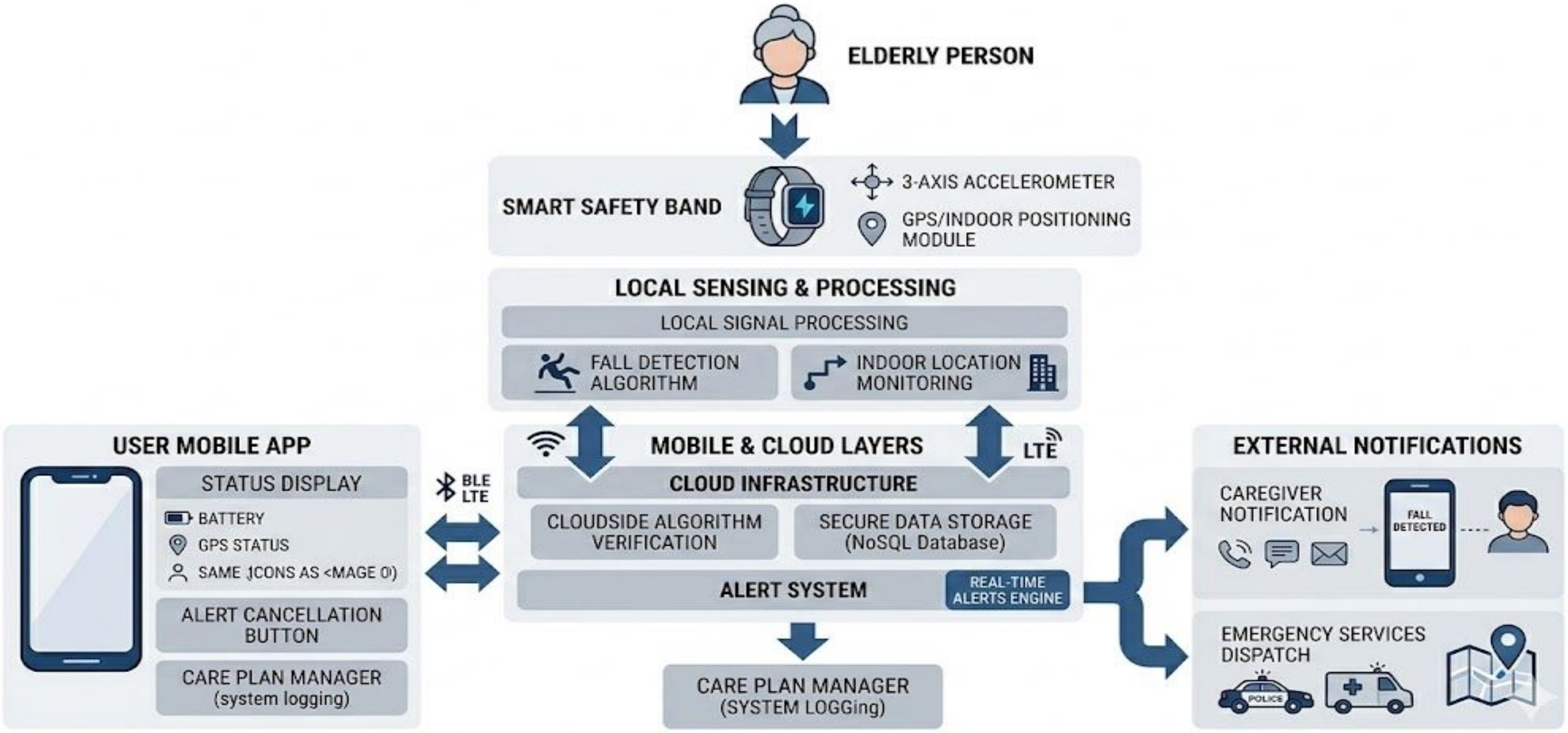

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WEARABLE SAFETY DEVICE FOR ELDERLY

Multi-View Technical Illustration

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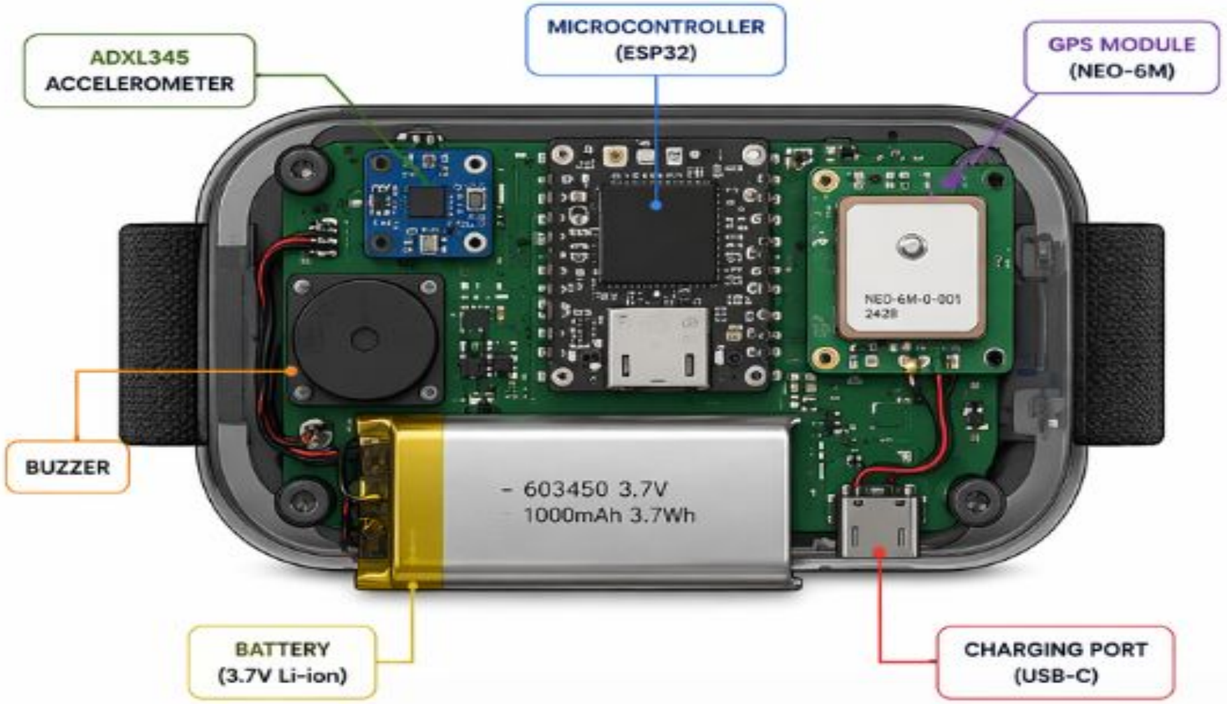
TOP VIEW (EXTERIOR)

SIDE VIEW

FRONT VIEW

INTERNAL LAYOUT (TOP VIEW – WITH SEMI-TRANSPARENT COVER)



COMPONENT LEGEND

- Microcontroller (ESP32)
- ADXL345 Accelerometer
- GPS Module (Neo-6M)
- Buzzer
- 3.7V Li-ion Battery
- USB-C Charging Port

KEY FEATURES

- Fall Detection using ADXL345
- Real-time Location Tracking (GPS)
- Instant Alert with Buzzer & SOS
- Compact, Lightweight & Wearable
- Long Battery Life

EXPLODED ISOMETRIC VIEW

TEAM NAME

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Estimated Cost

Component	Estimated Price (INR)
ESP32 Module (e.g., ESP-WROOM-32)	₹140 - ₹190
GSM Module (eg SIM800L or SIM800C)	₹140 - ₹200
Relay Module (eg Ra-01 or Ra-02)	₹50 - ₹ 80
GPS Module (eg Neo-6M)	₹245 - ₹300
Mini 3.7V Li-ion Battery	₹55 - ₹75
Custom ABS Plastic Enclosure	₹50 - ₹100
Total Raw BOM Cost	₹680 ₹945

1. Indoor Location (No GPS):
Uses step tracking and activity patterns for approximate positioning.

2. Alternatives for Indoor Positioning:
BLE Beacons (room-level accuracy)
Wi-Fi based positioning

3. Sensor Options:
ADXL345 (fall detection)
MPU6050 (advanced motion tracking)

4. Communication Options:
GSM/SMS (backup alerts)
LoRa (future scope for remote areas)

5. Navigation for Blind:
Audio guidance (voice instructions)
Vibration alerts (haptic feedback)

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Event Classification

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A1-C3 Emergency Response Matrix

	Class A: Active (Within City)	Class B: Bad (Poor Network)	Class C: Cut-off (Dead Zone)
Tier 1: Critical (Severe Injury)	**[A1] Hospital & Police** Auto-dials 108 & 100. Sends instant SMS with medical details & live GPS.	**[B1] Hospital & Police** Fires SMS with Medical Details & Location. Activates audio beacon.	**[C1] Hospital & Police** Initiates V2V (Vehicle-to-Vehicle) wireless signal relay flagging a critical emergency.
Tier 2: Urgent (Breakdown)	**[A2] Towing & Hospital** Notifies Towing service via app. Prompts user to auto-dial 108 if needed.	**[B2] Towing & Hospital** Sends SMS with Car No, Name, and Location. Activates audio beacon.	**[C2] Towing** Initiates V2V signal relay to broadcast breakdown status to passing cars.
Tier 3: Routine (Minor Accident)	**[A3] Police (Optional)** App prompt: "Are you okay?". If no response, sends standard SMS.	**[B3] System Log** Queues an SMS with location data to send once connectivity improves.	**[C3] System Log** Logs the crash data via hardware sensors. No immediate relay unless manually triggered.

Legend: [X#] = Priority Code = Hospital (108) = Police (100) = Towing Service = System Log

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Tech Stack

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For SmartSOS

Firebase

MAPS

REST API

twilio

For Communication Integration

WiFi

GSM

LoRa

Machine Learning

Accelerometer Data

Open-Source Large Language Models

LLMs

Regression Algorithms for Score Calculations

For Hardware

ESP32-S3-PICO-1

GPS

For Backend

node JS

Express JS

For Application

For Database

MongoDB

Live Demo

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9:26 28%

Smart Emergency Response

Crash detection active • Alerts auto-trigger

Monitoring Status

MONITORING ON

Crash detection active. Alerts trigger automatically.

Simulate Crash (Test)

Quick Access

Hospitals | History | Settings

Home | Hospitals | History | Analytics | Settings

9:26 28%

Performance Analytics

Measurable impact metrics.

Severity Distribution

0	0	2
Critical (Red)	Urgent (Yellow)	Stable (Green)

Avg CSI: 6.3

Measurable Impact (PDF)

Auto detection saves

2–3 min

No manual SOS required

Severity prioritization

3–5 min

Dispatch decisions

Pre-hospital preparation

5–8 min

Before arrival

Total potential reduction

10–15 min

Emergency cycle

Home | Hospitals | History | Analytics | Settings

THANK YOU